Kernix 0.13.0 / Performance brief

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

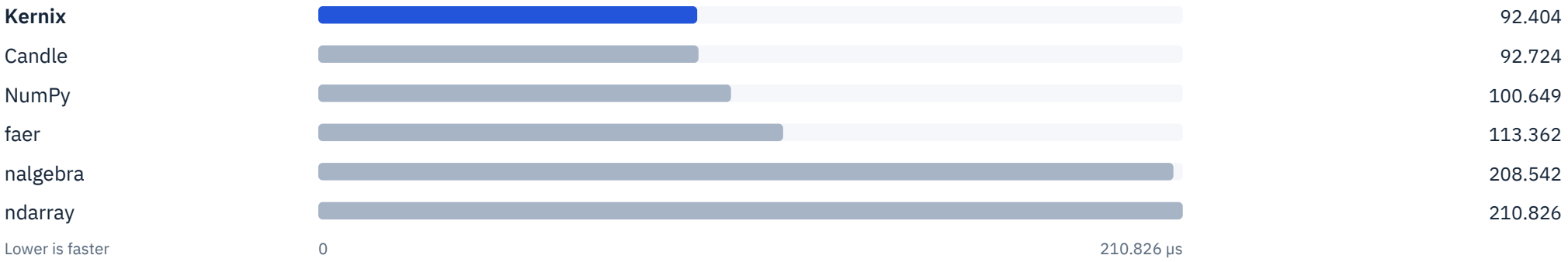
51 workloads

175 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.13.0 / Sparse matrices & memory

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Sparse matrix × vector

Per-call median

2048 × 2048 · five diagonals · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Sparse × dense matrix

Per-call median

512 × 512 · 32 RHS · five diagonals · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Kernix 0.13.0 / Sparse matrices & memory

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Sparse × dense matrix

Per-call median

512 × 512 · 32 RHS · five diagonals · f32 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



Sparse × sparse matrix

Per-call median

256 × 256 · five diagonals each · f64 · 1 worker · five diagonals · prepared operands · owned output · finite contracts differ



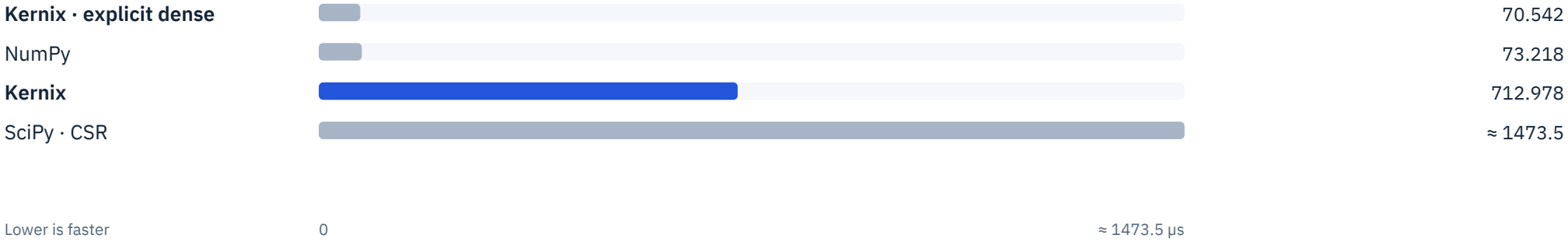
Kernix 0.13.0 / Sparse matrices & memory

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Automatic CSR construction

Per-call median

512 × 512 · clone + scan + compress · f64 · Construction · dense copy or compression · preparation costs included



Kernix 0.13.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication · transposed view

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · borrowed transpose · allocated output · independent operands



Matrix multiplication · transposed view

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · borrowed transpose · allocated output · independent operands



Kernix 0.13.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



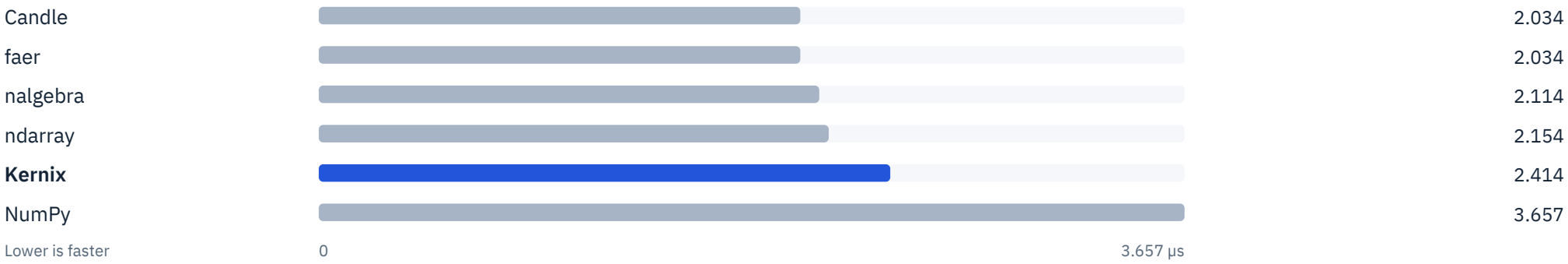
Kernix 0.13.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

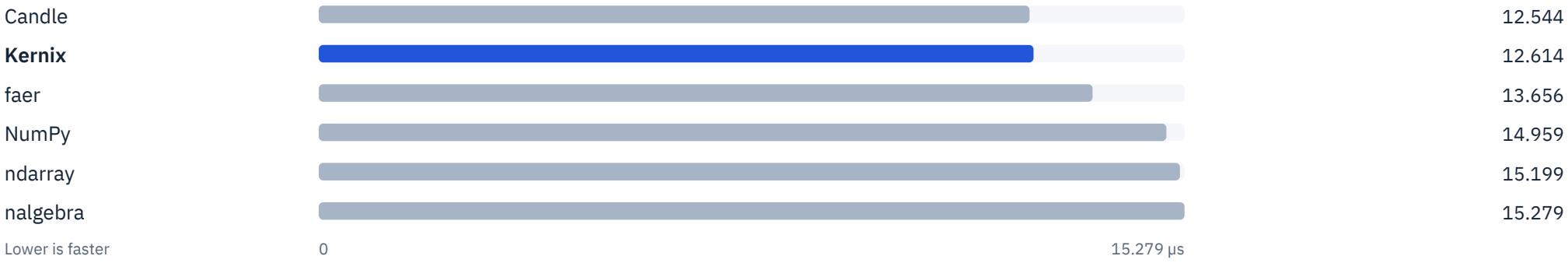
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



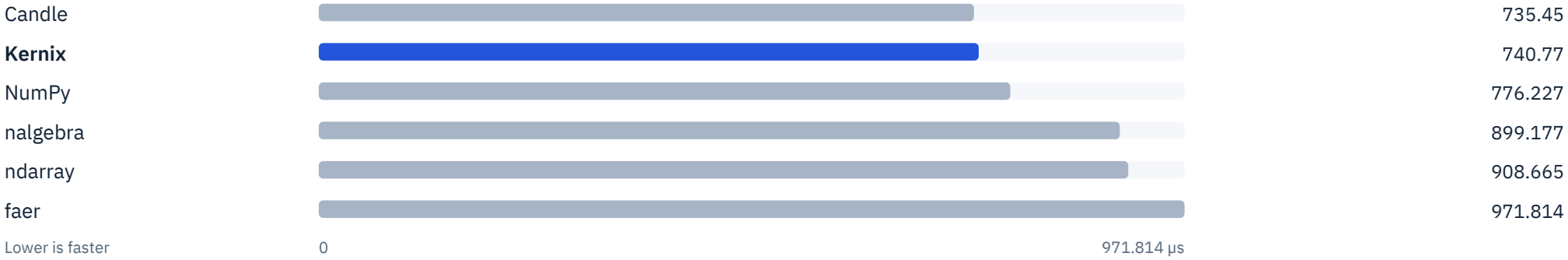
Kernix 0.13.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

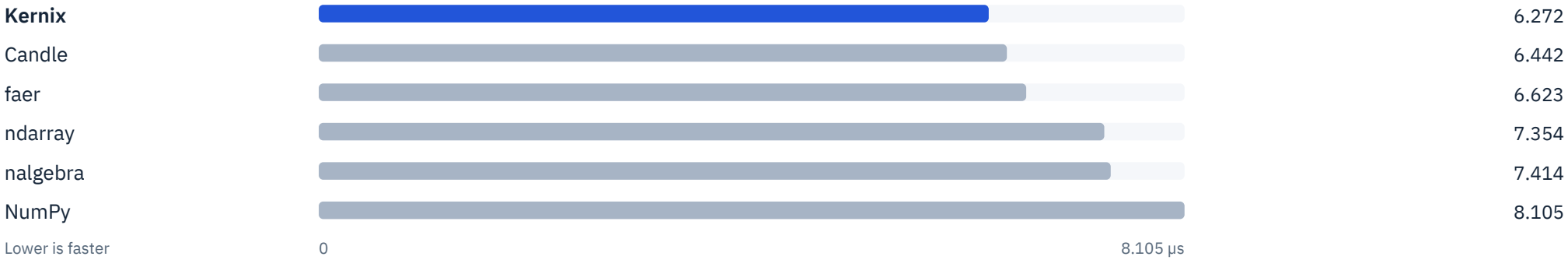
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



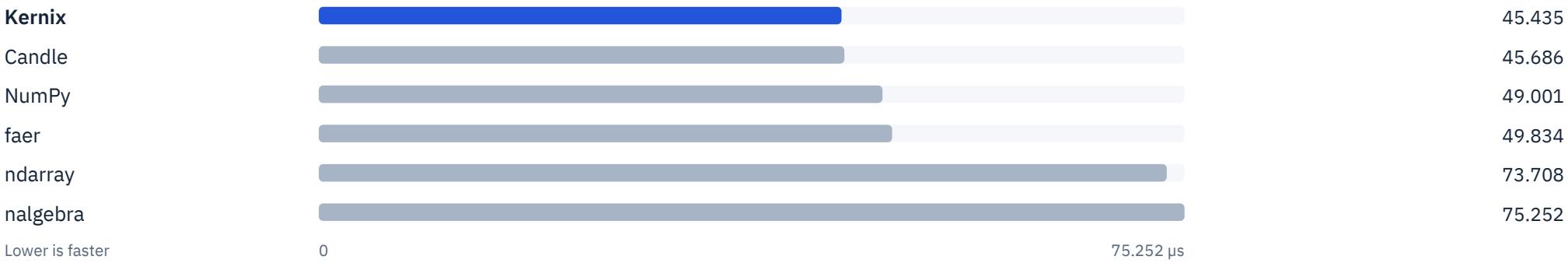
Kernix 0.13.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

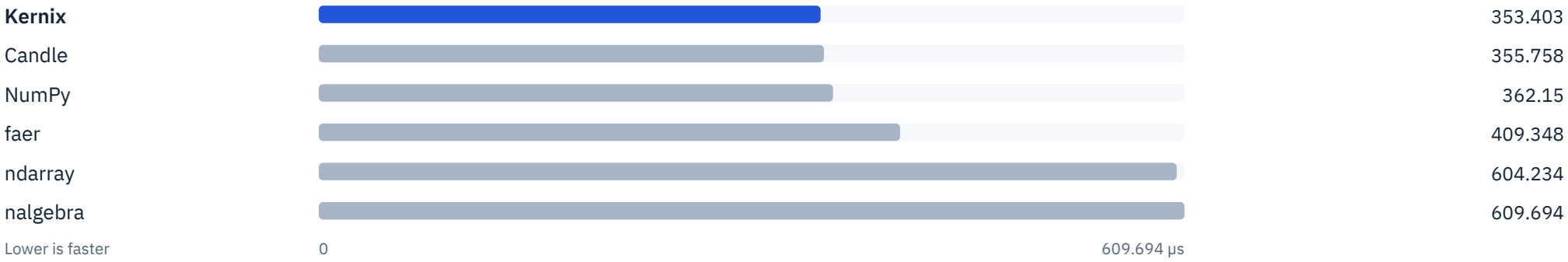
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



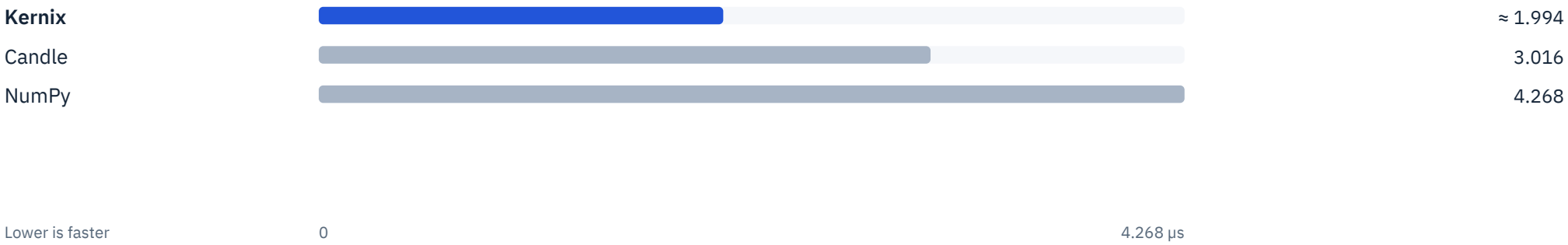
Kernix 0.13.0 / Matrix multiplication

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

Per-call median

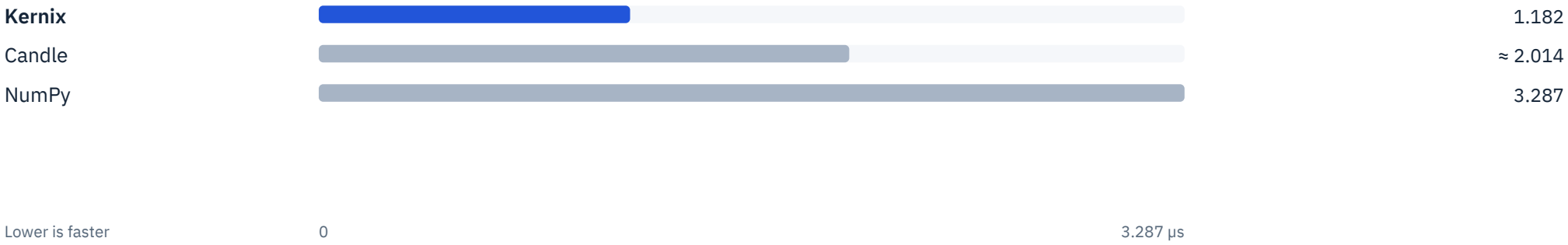
4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f32 · 1 worker · allocated output · broadcast batches



Kernix 0.13.0 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

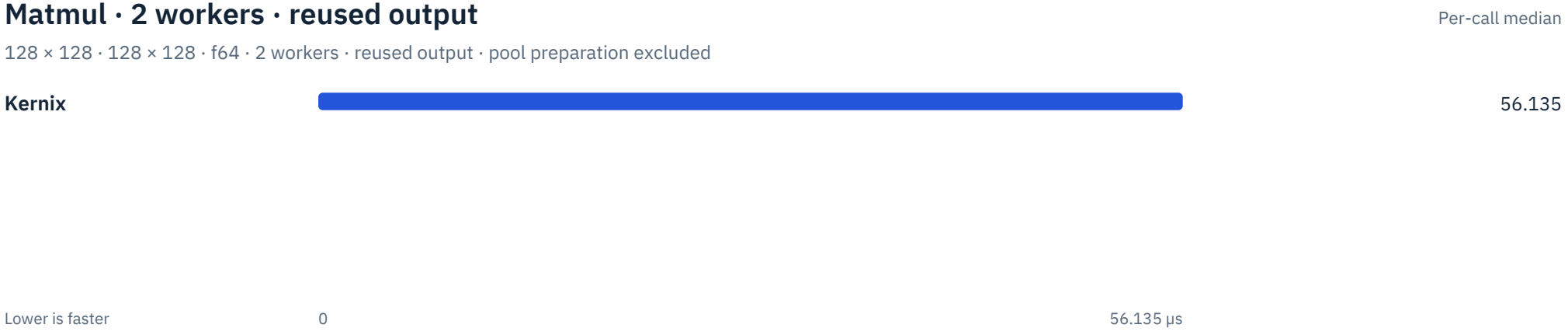
Matmul · 2 workers · reused output

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.13.0 / Explicit parallel execution

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



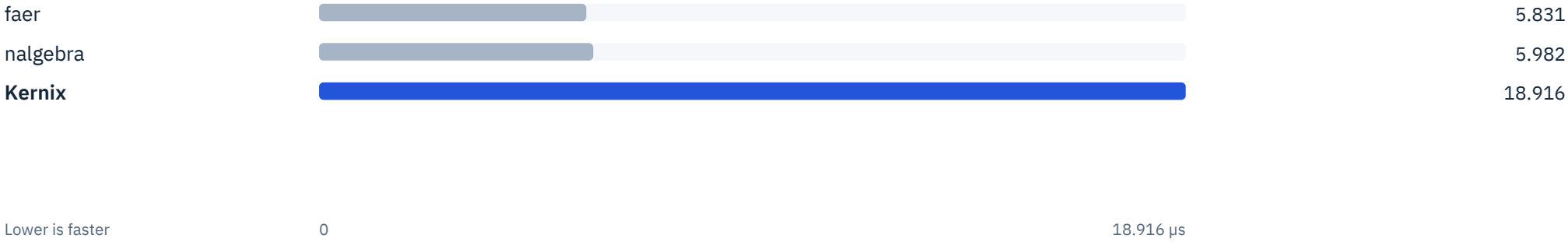
Kernix 0.13.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

128 × 128, 1 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Cholesky solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



Kernix 0.13.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky solve · reused factor

Per-call median

512 × 512, 128 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



LU solve · reused factor

Per-call median

256 × 256, 64 RHS · f64 · 1 worker · prepared reusable factor · owned output · factorization excluded



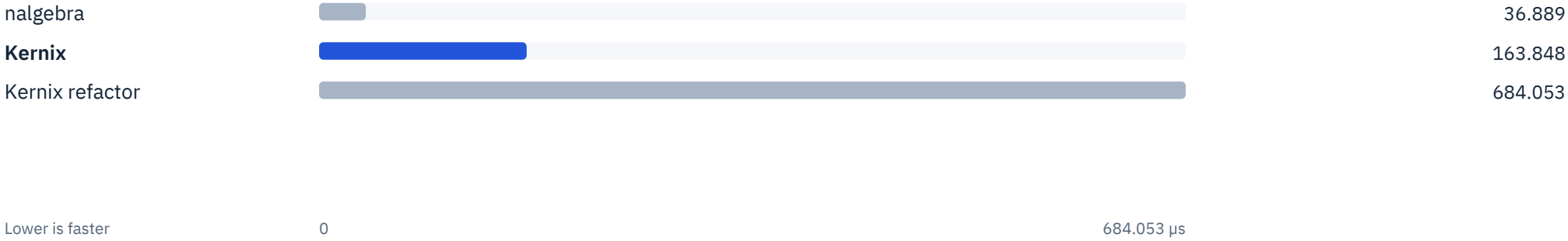
Kernix 0.13.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky rank-one update

Per-call median

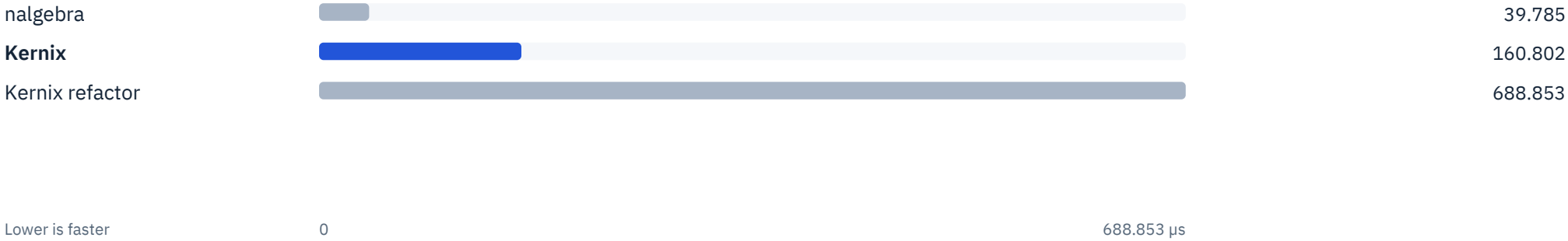
256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



Cholesky rank-one downdate

Per-call median

256 × 256 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix



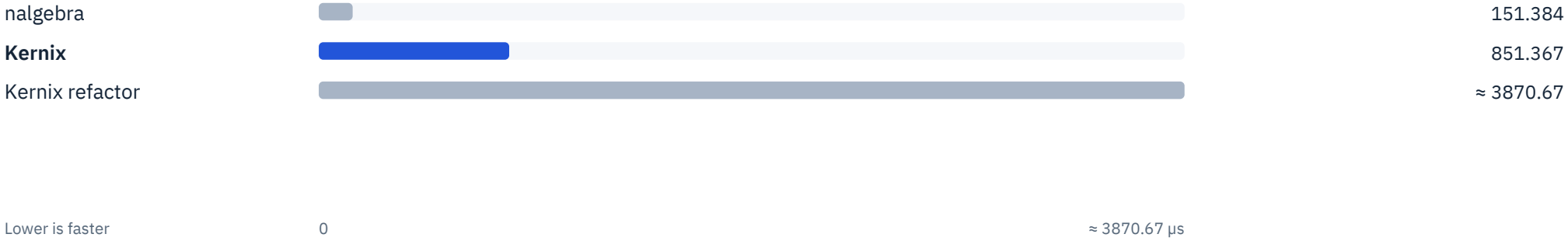
Kernix 0.13.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky rank-one update

512 × 512 · f64 · Prepared factors · owned output · refactor uses preformed modified matrix

Per-call median



Thin QR

512 × 128 · f64 · 1 worker · allocated output

Per-call median



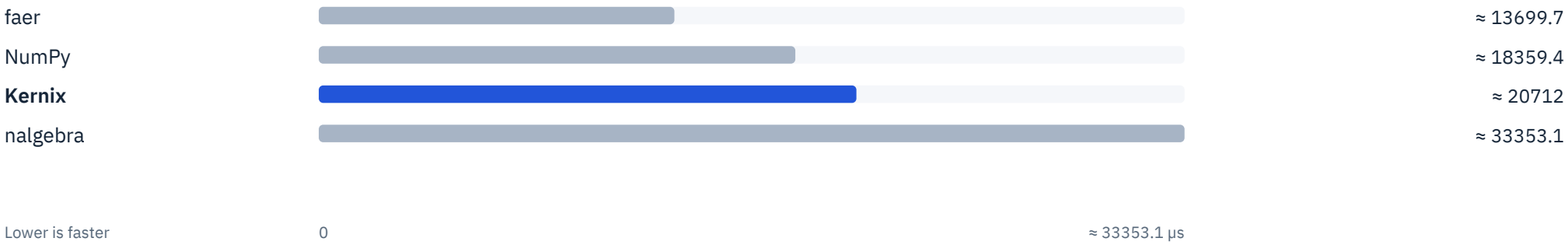
Kernix 0.13.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

1024 × 256 · f64 · 1 worker · allocated output

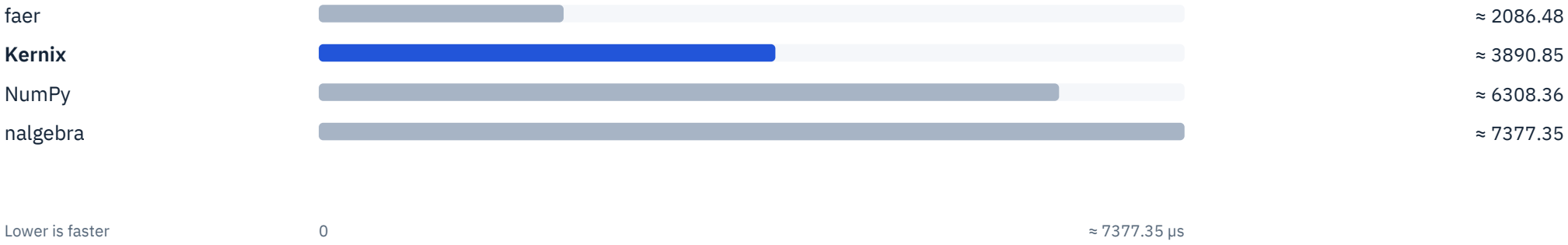
Per-call median



Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output

Per-call median



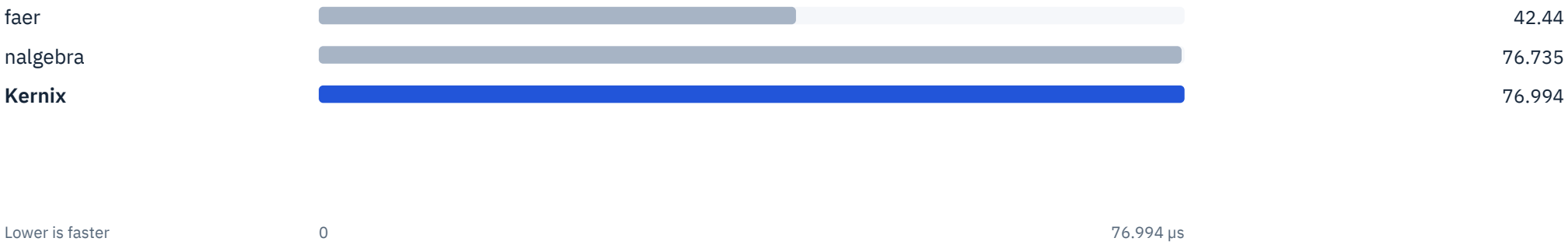
Kernix 0.13.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

Per-call median



Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

Per-call median



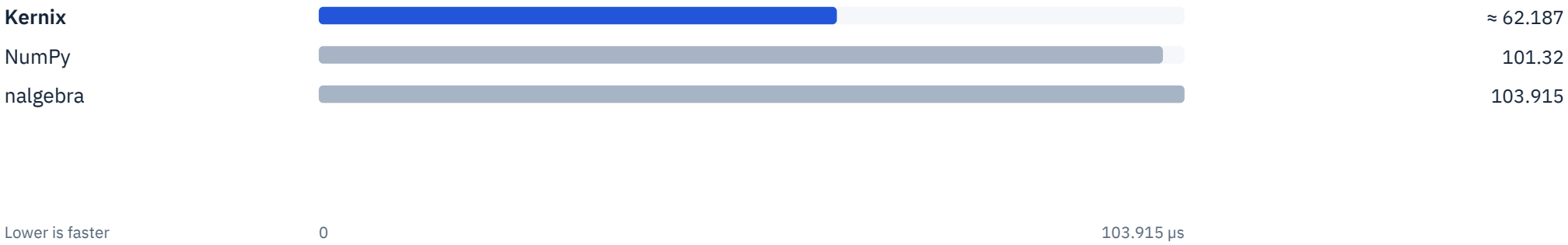
Kernix 0.13.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

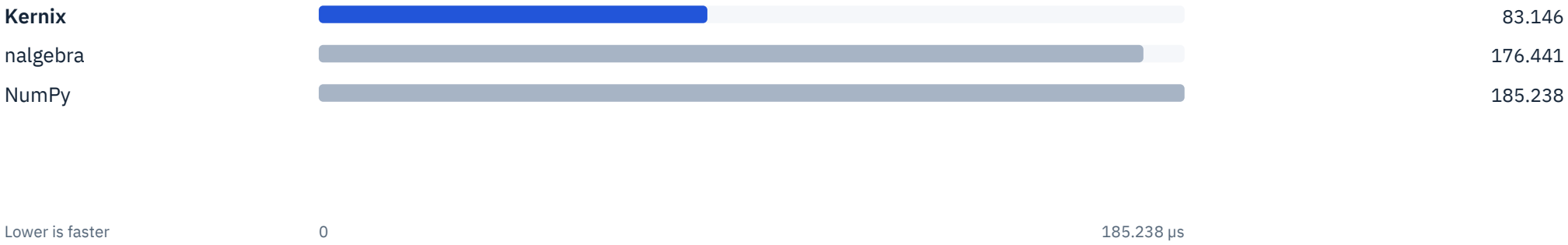
64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Kernix 0.13.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output



Kernix 0.13.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

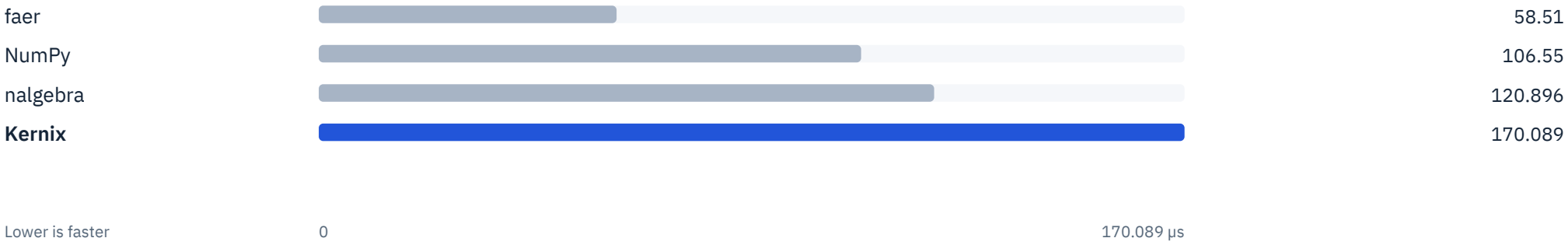
64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output



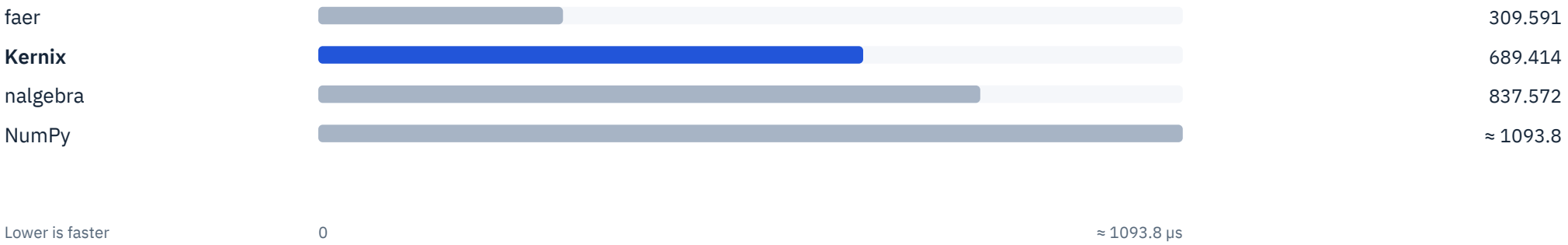
Kernix 0.13.0 / Decompositions & solvers

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.13.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

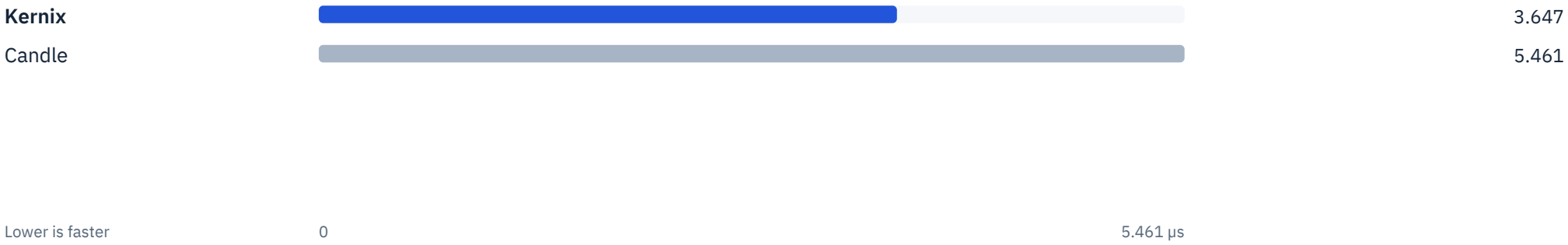
sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.13.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model inference

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented

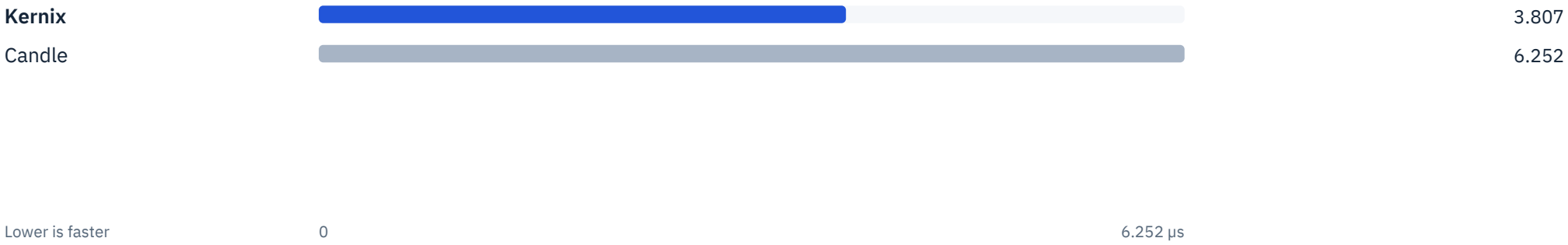
Per-call median



Model recording + forward

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented

Per-call median



Kernix 0.13.0 / Training & autodiff

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



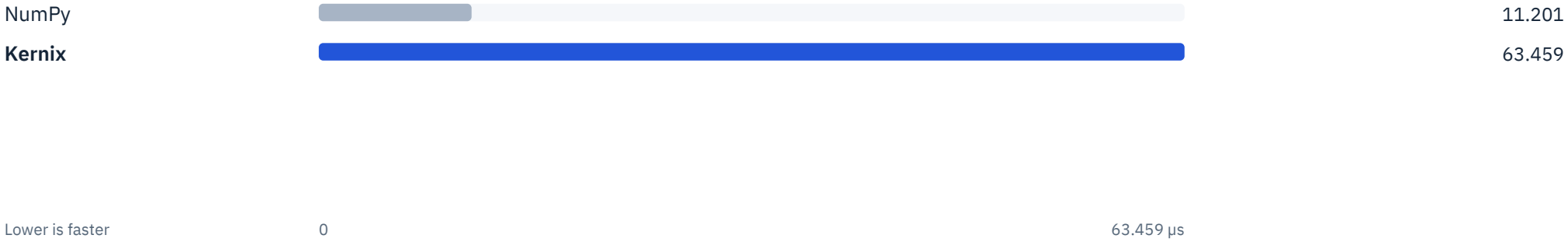
Kernix 0.13.0 / Tensor contractions

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.13.0 / Statistics & activations

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output

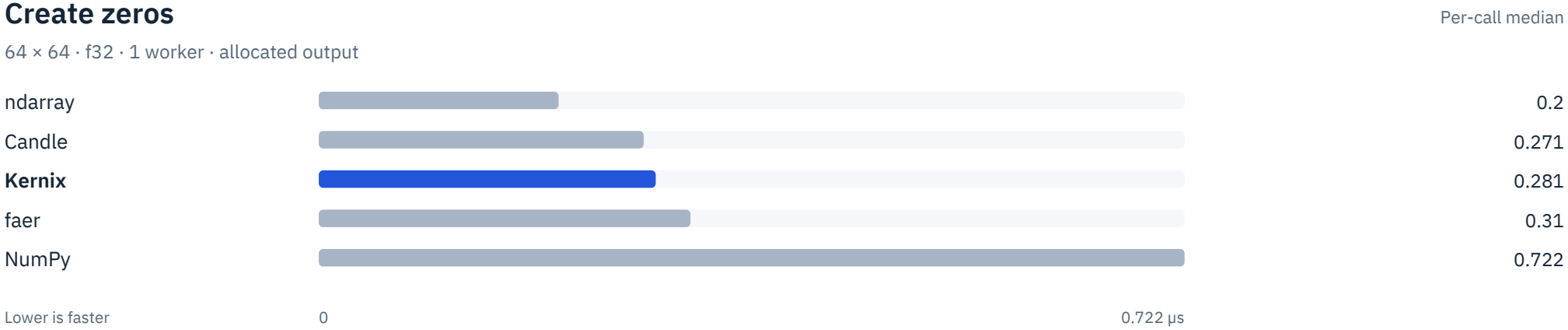


Kernix 0.13.0 / Tensor construction

AMD EPYC 7763 64-Core Processor · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

64 × 64 · f32 · 1 worker · allocated output



Create ones

64 × 64 · f32 · 1 worker · allocated output



Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky solve · reused factor

128 × 128, 1 RHS · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.999471067386015	0.9973565952947396 – 1.0020628371945415

Cholesky solve · reused factor

256 × 256, 64 RHS · f64

Observed time increase



CURRENT / BASELINE	95% INTERVAL
1.083568245363017	1.0802557742871344 – 1.08517793042905

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

LU solve · reused factor

256 × 256, 64 RHS · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.996840299031319	0.9959151607706278 – 0.9975877328672669

Thin QR

512 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9967563158814609	0.9918963979065318 – 1.0046826330122811

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Thin QR

1024 × 256 · f64

Observed time reduction



CURRENT / BASELINE	**0.9871204439911031**	95% INTERVAL	0.9779744047084407 – 0.9891396504940139

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	**0.9998129801151108**	95% INTERVAL	0.9957627512381872 – 1.0028547266517467

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication · transposed view

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0028165236051503	0.9960981496379726 – 1.0098207241763921

Thin QR

256 × 64 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9987499230413007	0.9846317345901487 – 1.0057244869354816

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

32 × 32 · 32 × 32 · f64

Timing direction unclear

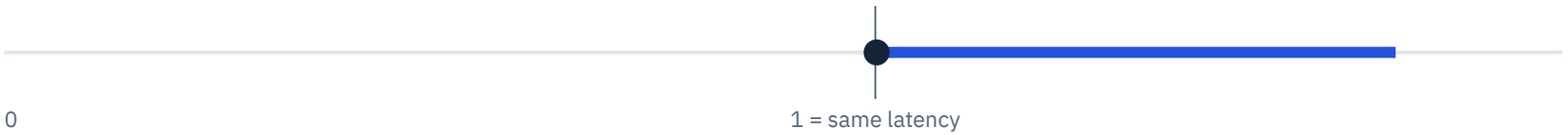


CURRENT / BASELINE	95% INTERVAL
0.9963023829087921	0.9837728194726166 – 1.0041597337770383

Matrix multiplication

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0013946850606525	0.9919242970052153 – 1.597163381838276

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.000878445152081	0.9969261170271161 – 1.0062039122179538

Matrix multiplication

256 × 256 · 256 × 256 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.9981894802135081	0.9808812825940344 – 0.9992079677625157

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0007649743733584	0.996856598178352 – 1.001591439699754

Record + backward

sum(A · W), 32 × 16 · 16 × 8 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9972580202906498	0.9864755206924534 – 1.0137098985467508

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0011809544258952	0.994669793721017 – 1.0053511348464619

Independent matrix operands

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.000462300495322	0.9910861814070788 – 1.0065968862696808

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0049407114624507	0.9856648541769648 – 1.009930486593843

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9916805324459235	0.9916107382550335 – 1.007607776838546

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.000831230693046	0.9978279884237653 – 1.004733574143589

Cholesky factorization

256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0035265993610598	0.9962365833107335 – 1.0075267678861823

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Observed time increase



CURRENT / BASELINE	95% INTERVAL
1.009350517607689	1.0043837900899306 – 1.0151540833691663

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
0.9973336177474402	0.9863506001487095 – 1.0044559308439533

Version comparison / paired processes

AMD EPYC 7763 64-Core Processor · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.00182105104314	0.9877555854589073 – 1.0068579077147009

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Timing direction unclear



CURRENT / BASELINE	95% INTERVAL
1.0022567541427556	0.9981015509363537 – 1.004204377687913

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Sparse matrix × vector / faer 2048 × 2048 · five diagonals · f64	7244	7.244
Sparse matrix × vector / SciPy · CSR 2048 × 2048 · five diagonals · f64	10430	10.43
Sparse matrix × vector / NumPy 2048 × 2048 · five diagonals · f64	652023	652.023
Sparse matrix × vector / Kernix · explicit dense 2048 × 2048 · five diagonals · f64	2617301	2617.301
Sparse matrix × vector / Kernix 2048 × 2048 · five diagonals · f64	18394	18.394
Sparse × dense matrix / SciPy · CSR 512 × 512 · 32 RHS · five diagonals · f64	27050	27.05
Sparse × dense matrix / Kernix · explicit dense 512 × 512 · 32 RHS · five diagonals · f64	611327	611.327
Sparse × dense matrix / NumPy 512 × 512 · 32 RHS · five diagonals · f64	482964	482.964
Sparse × dense matrix / faer 512 × 512 · 32 RHS · five diagonals · f64	66495	66.495
Sparse × dense matrix / Kernix 512 × 512 · 32 RHS · five diagonals · f64	34334	34.334

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Sparse × dense matrix / SciPy · CSR 512 × 512 · 32 RHS · five diagonals · f32	18585	18.585
Sparse × dense matrix / Kernix 512 × 512 · 32 RHS · five diagonals · f32	20188	20.188
Sparse × dense matrix / Kernix · explicit dense 512 × 512 · 32 RHS · five diagonals · f32	257984	257.984
Sparse × dense matrix / NumPy 512 × 512 · 32 RHS · five diagonals · f32	205395	205.395
Sparse × dense matrix / faer 512 × 512 · 32 RHS · five diagonals · f32	61334.99999999999	61.33499999999999
Sparse × sparse matrix / NumPy 256 × 256 · five diagonals each · f64	769644	769.644
Sparse × sparse matrix / SciPy · CSR 256 × 256 · five diagonals each · f64	67447	67.447
Sparse × sparse matrix / Kernix · explicit dense 256 × 256 · five diagonals each · f64	735439	735.439
Sparse × sparse matrix / Kernix 256 × 256 · five diagonals each · f64	166673	166.673
Automatic CSR construction / Kernix 512 × 512 · clone + scan + compress · f64	712978	712.978

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Automatic CSR construction / SciPy · CSR 512 × 512 · clone + scan + compress · f64	1473505	1473.505
Automatic CSR construction / Kernix · explicit dense 512 × 512 · clone + scan + compress · f64	70542	70.542
Automatic CSR construction / NumPy 512 × 512 · clone + scan + compress · f64	73218	73.218
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f64	101301	101.301
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f64	214931	214.931
Matrix multiplication · transposed view / Kernix 128 × 128 · 128 × 128 · f32	74360	74.36
Matrix multiplication · transposed view / NumPy 128 × 128 · 128 × 128 · f32	50655	50.655
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	93435	93.435
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	100458	100.458
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	49152	49.152

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	45695	45.695
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	2414	2.414
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	2034	2.034
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	3657	3.657
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	2154	2.154
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	2034	2.034
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	2114	2.114
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	15199	15.199
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	15279	15.279
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	12614	12.614

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	13656	13.656
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	12544	12.544
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	14959	14.959
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	210826	210.826
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	100649	100.649
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	113362	113.362
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	92404	92.404
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	208542	208.542
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	92724	92.724
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	740770	740.77

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	908665	908.665
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	776227	776.227
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	971814	971.814
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	735450	735.45
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	899177	899.177
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	8105	8.105
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	6272	6.272
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	6623	6.623
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	6442	6.442
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	7354	7.354

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	7414	7.414
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	75252	75.252
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	73708	73.708
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	45686	45.686
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	45435	45.435
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	49834	49.834
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	49001	49.001
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	604234	604.234
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	409348	409.348
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	609694	609.694

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	355758	355.758
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	362150	362.15
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	353403	353.403
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	3016	3.016
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	4268	4.268
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	1993.9999999999998	1.9939999999999998
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f32	3287	3.287
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f32	2014.0000000000002	2.0140000000000002
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f32	1182	1.182
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	15750	15.75

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	56135	56.135
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	393488	393.488
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	31218	31.218
Cholesky solve · reused factor / Kernix 128 × 128, 1 RHS · f64	18916	18.916
Cholesky solve · reused factor / faer 128 × 128, 1 RHS · f64	5831	5.831
Cholesky solve · reused factor / nalgebra 128 × 128, 1 RHS · f64	5982	5.982
Cholesky solve · reused factor / faer 256 × 256, 64 RHS · f64	275997	275.997
Cholesky solve · reused factor / Kernix 256 × 256, 64 RHS · f64	1322281	1322.281
Cholesky solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	1276014	1276.014
Cholesky solve · reused factor / faer 512 × 512, 128 RHS · f64	2212311	2212.311

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky solve · reused factor / nalgebra 512 × 512, 128 RHS · f64	9838110	9838.11
Cholesky solve · reused factor / Kernix 512 × 512, 128 RHS · f64	10263337	10263.337
LU solve · reused factor / faer 256 × 256, 64 RHS · f64	349305	349.305
LU solve · reused factor / Kernix 256 × 256, 64 RHS · f64	1176307	1176.307
LU solve · reused factor / nalgebra 256 × 256, 64 RHS · f64	1116935	1116.935
Cholesky rank-one update / Kernix 256 × 256 · f64	163848	163.848
Cholesky rank-one update / nalgebra 256 × 256 · f64	36889	36.889
Cholesky rank-one update / Kernix refactor 256 × 256 · f64	684053	684.053
Cholesky rank-one downdate / nalgebra 256 × 256 · f64	39785	39.785
Cholesky rank-one downdate / Kernix refactor 256 × 256 · f64	688853	688.853

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky rank-one downdate / Kernix 256 × 256 · f64	160802	160.802
Cholesky rank-one update / nalgebra 512 × 512 · f64	151384	151.384
Cholesky rank-one update / Kernix 512 × 512 · f64	851367	851.367
Cholesky rank-one update / Kernix refactor 512 × 512 · f64	3870674	3870.674
Thin QR / NumPy 512 × 128 · f64	3455533	3455.533
Thin QR / Kernix 512 × 128 · f64	3748874	3748.874
Thin QR / nalgebra 512 × 128 · f64	4449678	4449.678
Thin QR / faer 512 × 128 · f64	3153292	3153.292
Thin QR / nalgebra 1024 × 256 · f64	33353110.999999996	33353.110999999996
Thin QR / faer 1024 × 256 · f64	13699713	13699.713

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / Kernix 1024 × 256 · f64	20711954	20711.954
Thin QR / NumPy 1024 × 256 · f64	18359428	18359.428
Cholesky factorization / Kernix 512 × 512 · f64	3890852	3890.852
Cholesky factorization / NumPy 512 × 512 · f64	6308358	6308.358
Cholesky factorization / nalgebra 512 × 512 · f64	7377353	7377.353
Cholesky factorization / faer 512 × 512 · f64	2086485	2086.485
Column-pivoted QR / nalgebra 96 × 32 · f64	76735	76.735
Column-pivoted QR / Kernix 96 × 32 · f64	76994	76.994
Column-pivoted QR / faer 96 × 32 · f64	42440	42.44
Column-pivoted QR / Kernix 256 × 64 · f64	796976	796.976

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / nalgebra 256 × 64 · f64	755348	755.348
Column-pivoted QR / faer 256 × 64 · f64	381826	381.826
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	103915	103.915
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	62187.00000000001	62.18700000000001
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	101320	101.32
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	83146	83.146
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	176441	176.441
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	185238	185.238
Thin QR / Kernix 96 × 32 · f64	66866	66.866
Thin QR / nalgebra 96 × 32 · f64	57949	57.949

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / NumPy 96 × 32 · f64	69130	69.13
Thin QR / faer 96 × 32 · f64	54031	54.031
Thin QR / faer 256 × 64 · f64	422122	422.122
Thin QR / Kernix 256 × 64 · f64	745830	745.83
Thin QR / NumPy 256 × 64 · f64	542488	542.488
Thin QR / nalgebra 256 × 64 · f64	554731	554.731
Cholesky factorization / faer 64 × 64 · f64	14377	14.377
Cholesky factorization / Kernix 64 × 64 · f64	24977	24.977
Cholesky factorization / nalgebra 64 × 64 · f64	20849	20.849
Cholesky factorization / NumPy 64 × 64 · f64	18164	18.164

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 128 × 128 · f64	170089	170.089
Cholesky factorization / faer 128 × 128 · f64	58510	58.51
Cholesky factorization / NumPy 128 × 128 · f64	106550	106.55
Cholesky factorization / nalgebra 128 × 128 · f64	120896	120.896
Cholesky factorization / nalgebra 256 × 256 · f64	837572	837.572
Cholesky factorization / faer 256 × 256 · f64	309591	309.591
Cholesky factorization / NumPy 256 × 256 · f64	1093802	1093.802
Cholesky factorization / Kernix 256 × 256 · f64	689414	689.414
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	4558	4.558
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	10309	10.309

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	3647	3.647
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	5461	5.461
Model inference / Candle batched normalized/sliced MLP · f64	4399	4.399
Model inference / Kernix batched normalized/sliced MLP · f64	2985	2.985
Model recording + forward / Candle batched normalized/sliced MLP · f64	6252	6.252
Model recording + forward / Kernix batched normalized/sliced MLP · f64	3807	3.807
Model backward / Kernix prepared graph → fresh gradients · f64	4718	4.718
Model backward / Candle prepared graph → fresh gradients · f64	16822	16.822
Parameter update / Kernix transactional SGD, fixed gradients · f64	792	0.792
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	11201	11.201

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	63459	63.459
Row softmax / ndarray 256 × 64 · f64	80892	80.892
Row softmax / NumPy 256 × 64 · f64	119875	119.875
Row softmax / Candle 256 × 64 · f64	84519	84.519
Row softmax / Kernix 256 × 64 · f64	94247	94.247
Create zeros / faer 64 × 64 · f32	310	0.31
Create zeros / Candle 64 × 64 · f32	271	0.271
Create zeros / ndarray 64 × 64 · f32	200	0.2
Create zeros / Kernix 64 × 64 · f32	281	0.281
Create zeros / NumPy 64 × 64 · f32	722	0.722

Recorded precision / exact latency table

AMD EPYC 7763 64-Core Processor · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create ones / faer 64 × 64 · f32	390	0.39
Create ones / ndarray 64 × 64 · f32	351	0.351
Create ones / Candle 64 × 64 · f32	441	0.441
Create ones / NumPy 64 × 64 · f32	2034	2.034
Create ones / Kernix 64 × 64 · f32	431	0.431